

An Embedded Gate Driver Architecture for Scalable Large-Area OLED Displays

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Abstract

Bezel-located gate drivers in large-area OLEDs suffer from timing degradation, causing non-uniform luminance. To address this, we propose a scalable architecture that distributes driver units within the active array to segment propagation paths. A 27-inch prototype demonstrated uniform falling and rising times of approximately 2 μs and 1 μs across the panel, ensuring high image quality.

Author Keywords

Active-matrix organic light-emitting diode (AMOLED); automotive displays; embedded gate driver; large-area display; resistive-capacitive (RC) delay.

1. Introduction

Automotive displays are rapidly evolving into large, dashboard-scale interfaces that enable seamless pillar-to-pillar cockpit designs and unified human-machine interfaces. Active-matrix organic light-emitting diode (AMOLED) technology is increasingly adopted in premium automotive large-area and free-form displays due to its high performance and design flexibility. In particular, low-temperature polycrystalline silicon (LTPS) thin-film transistor (TFT)-based AMOLED panels are widely adopted in automotive applications owing to their layout flexibility and compatibility with large-area integration [1]. While conventional vehicles employed multiple independent panels, recent designs increasingly integrate the instrument cluster, center information display, and co-driver display into a single, continuous OLED panel, resulting in a display size exceeding 50 inches, as illustrated in Figure 1.

However, as displays scale to larger sizes and adopt free-form layouts, conventional bezel-located gate driver architectures encounter scalability constraints. LTPS-based AMOLED panels require sufficient gate signal timing to ensure reliable threshold voltage (V_{TH}) compensation, and maintaining this timing becomes increasingly challenging as panel size increases [2], [3].

In conventional architectures, the scan signal is generated by a bezel-located gate driver-in-panel (GIP). By integrating gate driver circuits into the display panel, the GIP eliminates external driver ICs and packaging, thereby reducing cost and enhancing integration [4], [5], [6]. As panel dimensions expand, the extended routing length increases dependence on bezel-located driver placement, limiting scalable panel integration. This bezel-dependent architecture constrains layout scalability in large-area and free-form panels.

To enable scalable and layout-flexible large-area OLED integration, we introduce an embedded gate-driver architecture. The proposed structure redefines the driver placement strategy, allowing distributed signal injection points that enhance layout modularity and panel scalability. The effectiveness of the architecture was validated through simulations and prototype fabrication.



Figure 1. Dashboard-scale automotive display integrating multiple displays into a single panel.

2. Proposed Embedded Gate Driver Architecture

In a conventional bezel-located gate driver architecture, Figure 2 illustrates the size-dependent degradation of gate signal timing. Figure 2(a) shows that the center delay increases rapidly with panel size, particularly as the panel size approaches the 30-inch class. Figure 2(b) compares representative simulated gate signal waveforms at panel edge and center for 30-inch and 40-inch panels. For the 30-inch panel, the center delay remains measurable, and the delay increases substantially. In addition, the large timing mismatch between the panel edge and center significantly reduces the uniformity of V_{TH} compensation across the panel. For the 40-inch panel, a further increase in delay leads to severe waveform distortion, resulting in an invalid gate pulse at the panel center.

Because the signal propagation path is inherently determined by the panel geometry, bezel-based driver placement becomes increasingly inefficient as panel size increases. Consequently, this architecture-level constraint cannot be effectively addressed by device scaling or process optimization alone, as illustrated in Figure 3(a).

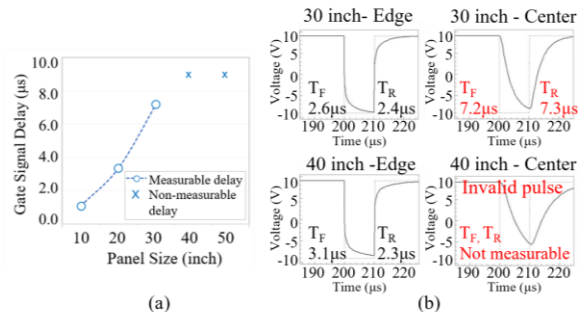


Figure 2. Simulation results in a conventional bezel-located gate driver architecture: (a) gate delay at the panel center as a function of panel size and (b) gate signal waveforms at the panel edge and center for 30-inch and 40-inch panels, simulated with an input gate pulse width of 10 μs .

To address this issue, the proposed architecture embeds gate driver units directly within the active array, as shown in Figure 3(b). The embedded gate driver architecture segments the gate signal propagation path into multiple regions by distributing driver units in the active array. Each region contains a single embedded gate driver unit. In this configuration, the effective signal propagation length is determined by the spacing between adjacent embedded gate driver units rather than by the overall panel size. As the panel size increases, additional gate driver units can be inserted at regular intervals to maintain a constant propagation length for each gate segment. As a result, the proposed embedded gate driver architecture reduces the dependence of gate signal timing on panel size, enabling scalable expansion to large and free-form displays.

The proposed architecture maintains the same timing sequence and circuit used in conventional gate drivers. Figure 4 presents a block-level comparison between the conventional bezel-located gate driver and the proposed embedded gate driver. As shown in Figure 4(a), the conventional gate driver circuit is located in the bezel region. In contrast, the proposed architecture embeds the same gate driver circuit within the active array, as illustrated in Figure 4(b), without modifying the original circuit. The proposed architecture requires no modification to existing gate driver circuits or LTPS TFT-based AMOLED processes, enabling straightforward integration into current manufacturing platforms.

By deploying multiple gate driver units along a single gate line, the effective signal propagation length is significantly shortened. The proposed cluster-based gate driver architecture represents a layout-level design approach that addresses RC delay, parasitic coupling, and scalability.

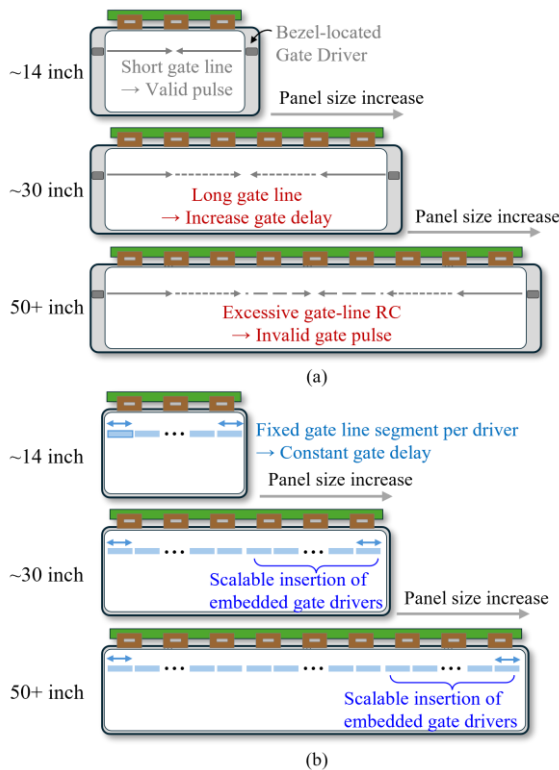


Figure 3. Conceptual comparison of gate driver architectures as a function of panel size (a) conventional bezel-located gate driver architecture and (b) proposed embedded gate driver architecture.

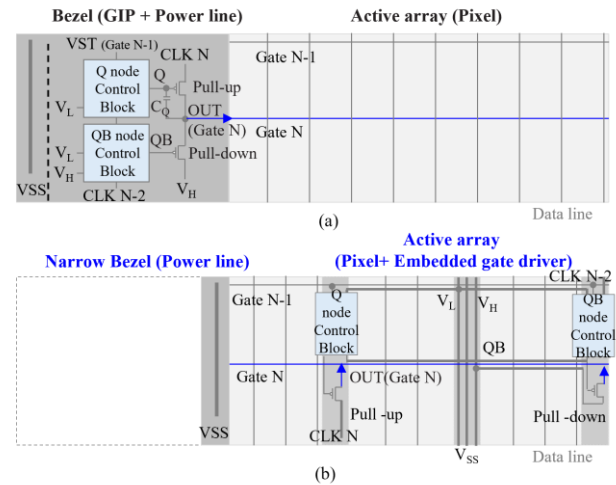


Figure 4. Block-level comparison of (a) a conventional bezel-located gate driver and (b) the proposed embedded gate driver integrated within the active array, enabling localized routing without circuit and process modification.

When gate drivers are implemented as individually distributed TFTs within the active array, increased routing complexity and longer interconnected nodes can introduce additional parasitic effects, which can degrade gate driver stability [7]. In contrast, integrating each gate driver as a compact cluster block shortens inter-node distances, thereby reducing parasitic effects. As a result, the proposed architecture ensures stable signal generation and reliable operation. Furthermore, embedding the gate drivers within the active array mitigates capacitive coupling between the gate-driver AC nodes and pixel circuits. This coupling is suppressed through electrical shielding using reference bias lines. This layout supports stable gate driver operation while enabling scalable expansion to large-area panels.

3. Prototype & Measurement Results

A 27-inch prototype panel was fabricated as a representative test vehicle, corresponding to the onset region where gate signal delay begins to significantly affect signal settling in conventional bezel-located gate driver architectures. While experimental validation was performed on the 27-inch prototype, the proposed embedded gate driver architecture is architecturally scalable to larger panel sizes.

The gate driver circuits were implemented using a standard commercial top-gate p-type LTPS process, without any circuit or process modifications. Embedded gate driver modules were uniformly distributed along each gate line to ensure consistent signal propagation. The gate driver circuits employed optimized device dimensions and capacitance values to ensure stable operation.

Figure 5(a) shows the embedded gate driver layout integrated within the active array. The cluster layout enables compact placement of the gate driver circuits while maintaining routing compatibility with the pixel-array layout and effective shielding. Figure 5(b) presents the superimposed measured scan-signal waveforms at the panel edge and center. Due to the uniform distribution of embedded gate driver modules along each gate line, the waveforms from the edge and center regions are well aligned and virtually indistinguishable, indicating consistent falling and

rising times of approximately 2 μs and 1 μs , respectively. In contrast to conventional bezel-located gate driver architectures, the waveforms confirm the elimination of signal distortion and propagation delay between the edge and center regions. These results demonstrate that the proposed embedded gate driver architecture effectively segments the signal-propagation length along the gate line, providing sufficient driving time for accurate V_{TH} compensation. Consequently, the stabilized gate timing enables uniform gate signal delivery, which is beneficial for achieving uniform luminance across the panel.

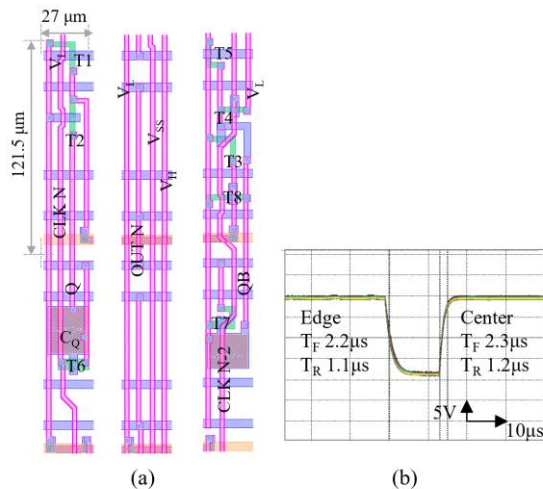


Figure 5. (a) Embedded gate-driver layout in the active array and (b) superimposed measured scan-signal waveforms at the panel edge and center, demonstrating negligible signal delay and uniform transition timing across the panel.

4. Conclusion

This work addressed the scalability limitation of conventional bezel-located gate driver architectures in large and free-form automotive OLED displays, where RC-induced delay degrades gate signal timing as panel size increases. To overcome this issue, an embedded gate driver architecture was proposed, in which multiple gate driver units are distributed within the active array. The proposed approach is fully compatible with existing gate driver circuits and LTPS-based AMOLED processes, enabling straightforward integration into current manufacturing platforms. A 27-inch prototype panel was fabricated to verify the proposed architecture, demonstrating uniform gate signal waveforms between the panel edge and center and stable gate driver operation. The results demonstrate architectural and system-level advantages of the proposed embedded gate driver scheme, supporting its applicability to large-area and free-form displays.

5. References

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